# Phase changes

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Account for energy transfer during a phase change without a temperature rise.

**SUCCESS CRITERIA - I can...**

1. Identify a constant-temperature phase-change interval.
2. Use Q=mL for that interval.
3. Separate warming energy from phase-change energy.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

Two objects at 280 K have different masses and materials. Must their internal energies be equal? Give two reasons.

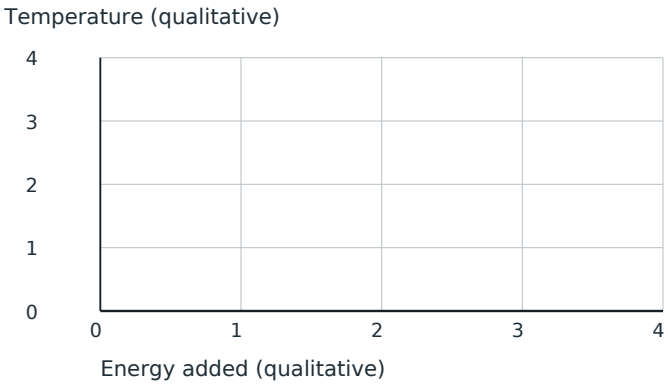
**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** Q = mL; Q = mc ΔT.
Use mL during a phase change at constant temperature, and mc ΔT outside that interval. L is latent heat.

**Worked reasoning:** Melting 2 kg of a substance with latent heat 100 J/kg requires 200 J. During ideal melting at fixed pressure, added energy changes phase without raising temperature.

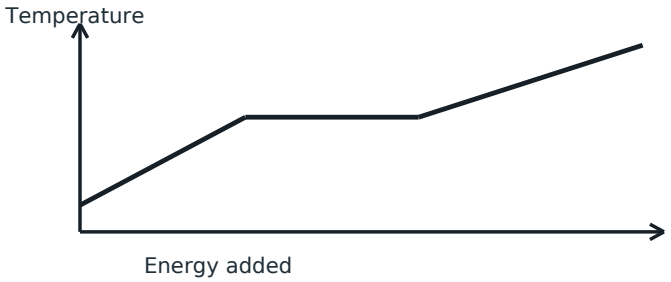
**Your turn:** Sketch temperature versus added energy through warming and melting. Mark the plateau.




**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

On the heating curve, identify the flat interval. Explain why added energy does not raise temperature during that interval.




The horizontal axis shows energy added. The vertical axis shows temperature.

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A beaker contains 0.10 kg of ice at 0°C. First melt it, then warm the water to 20°C. Use $L_f$ = 334,000 J/kg and $c_{water}$ = 4200 J/(kg·°C). Label the two stages, calculate each energy transfer and find the total.

| Stage | Equation | Energy (J) |
|---|---|---|
| Melt at 0°C | ____ | ____ |
| Warm 0→20°C | ____ | ____ |

**Marzano check - circle:** ① ② ③

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Continue down each column.

### 05 Explain your reasoning

A student uses mc ΔT for a melting interval with ΔT=0 and concludes no energy is required. Explain the missing term.

**Marzano check - circle:** ① ② ③

### 06 Review: connect an earlier idea

With right positive, a spring has k=200 N/m and displacement x=-0.06 m. Find signed spring force and direction.
Mark equilibrium, displacement and restoring-force direction.



Mark equilibrium and displacement x.

**Marzano check - circle:** ① ② ③

### 07 Exit check

A 0.20 kg vapor sample releases 45,000 J as it condenses to liquid at constant temperature. Label the outgoing energy arrow. Find its latent heat of condensation and state the sign of Q for the sample.

Initial T / phase: _____ Final T / phase: _____




Add an energy arrow; label mass, c or L as needed.

**Marzano check - circle:** ① ② ③

### 08 STRETCH: test the model

A 0.2 kg ice sample at 0°C melts and the water warms to 10°C. Use $L_f$=334000 J/kg and $c_{water}$=4200 J/(kg·°C). Find total heat. Label the energy arrow and identify what changes in the sample.

Initial T / phase: _____ Final T / phase: _____




Add an energy arrow; label mass, c or L as needed.

**Marzano check - circle:** ① ② ③